

Ex - 02

Task - 2 :

Proof:

Given the iterative definition of feedforward layer:

$$\tilde{z}_l = z_{l-1} \cdot B_l + b_l$$

Bias is not absorbed.

$$z_l = \phi_l(\tilde{z}_l)$$

If ϕ_l is identity function, then $z_l = \tilde{z}_l$

The layer simplifies to:

$$z_l = z_{l-1} \cdot B_l + b_l$$

Input layer $z_0 = X$

Layer 1: $z_1 = X \cdot B_1 + b_1$

Layer 2: $z_2 = z_1 \cdot B_2 + b_2$

Substituting z_1 into z_2 :

$$z_2 = (X \cdot B_1 + b_1) \cdot B_2 + b_2$$

$$z_2 = X \cdot (B_1 \cdot B_2) + (b_1 \cdot B_2 + b_2)$$

Comment: Induction would mean showing by induction, expanding this linear recursion up to the final output layer L yields:

$$z_L = X \cdot (B_1 B_2 \dots B_L) + (b_1 B_2 \dots B_L + b_2 B_3 \dots B_L + \dots + b_L)$$

neither is explicitly done. It is assumed, that it just works.

→ Because matrix multiplication is associative, the product of weight matrices & nested bias collapses into a simple matrix & a single vector respectively.

So, let

$$B' = B_1, B_2, \dots, B_L$$

$$b' = b_1, B_2, \dots, B_L + b_2, B_3, \dots, B_L + \dots + b_L$$

Substituting back into our final eqn we get,

$$Z_L = X \cdot B' + b'$$

- This final expression is mathematically and structurally identical to a standard 1-layer neural network using the weight matrix B' & bias vector b' .

$\therefore$ any feedforward network of depth $L > 1$ w/ linear identity activation functions is equivalent to a 1-layer neural network.

Proof by induction lacks the explicit induction steps but other than that it's fine.